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Diagnosis of supraharmonics-related problems based on the effects on electrical equipment

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ABSTRACT

Supraharmonics (SH) have proliferated in low-voltage (LV) and medium-voltage (MV) grids due to the increasing use of technologies emitting distortion in the range 2–150 kHz. Currently, no recommended practices to assess the effects of SH on the electrical system exist. The propagation of SH through LV and MV grids leads to interference with the elements for power delivery and end-user equipment, e.g., light flicker, aging of capacitors and cable terminations, audible noise, and interruption of electric vehicle (EV) charging. As such incidents happen more often, the need for guidelines that facilitate the diagnosis of problems related to SH arises. Different features of the SH distortion are responsible for different interferences. This article introduces guidelines for the evaluation of the impact of SH based on the morphology of the interference. The evaluation is performed with easy-to-use methods and formulas directly related to the characteristics of the SH distortion. The adverse effects of SH considered are audible noise, light flicker, tripping of LV residual current devices (RCDs), and MV cable termination failure. This work interests field engineers and researchers facing problems related to SH; it also serves as a guide for further research.

1. Introduction

Supraharmonics (SH) are current and voltage waveform distortion in the range 2 to 150 kHz. They can be created intentionally by power line communication (PLC) systems or unintentionally by power electronics converters. Driven by the need of increasing the efficiency of the power system, the shift of the power delivery paradigm into modern schemes incites the use of technologies that contribute to the proliferation of SH in electricity networks [1]. In the context of smart grid, smart metering and the monitoring of power quality utilize narrow-band PLC systems [2], and the installation of distributed generation together with the use of renewable energy sources is incited. Electrification of the transportation sector is ongoing in several countries [3,4]. In modern society, data management, cloud-based services, and the increasing use of artificial intelligence require the electricity supply to high-performance computing loads. The described activities involve the proliferation of loads with switched-mode power supplies and converter-based equipment using switching frequencies whose remnants appear as SH emitted into the grid [5]. The latter has led to more frequent reports of problems associated with SH, e.g., in CLC TR 50669 and [5–8].

Some examples of equipment that emit SH are: photovoltaic inverters, wind turbine converters, electric vehicle chargers, variable speed drives, LED lamps, air conditioning units, computer power supplies [1,4,5,7]. Ref. [9] concluded that SH are present in low-voltage (LV) grids in different parts of the world; the level of SH presents a daily pattern and is higher during the day; the frequency of the SH is also varying. The daily pattern depends on the location. Quantification of the voltage distortion over the whole SH range showed that the average values are between 0.09 and 0.56 % of the fundamental voltage though it can be as high as 12 % with a duration of less than one minute. In [4], SH emitted by electric vehicles were measured; the study reported SH between frequencies from 10 up to 60 kHz and magnitudes from 30 up to 1000 mA; a single device can emit SH at multiple frequencies. Ref. [10] showed levels of SH emitted by photovoltaic inverters in an MV installation with a magnitude of up to 2 % of the fundamental voltage. The emission coincided with the switching frequency of the inverters and its multiples, and it is present while power production takes place. In [11], a SH measurement campaign in an MV installation was performed; SH with magnitude up to 400 V at 2 kHz in a 70 kV busbar were reported. The emission is due to static frequency converters and the levels of SH depend on the configuration of the feeding network and the number of converters in operation. Ref. [12,13] showed the correlation of SH with

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the operation of renewable power plants in an MV network. Some components were shown to propagate over distances of about 16 km.

The study of SH as a power quality problem is relatively recent [7]. The lack of standardization and research on SH has left gaps in the methodology on how to assess SH distortion as a characteristic of the power quality. In the context of adverse effects of SH, the gaps that motivate this work are explained: 1) Although efforts to develop indices for SH have been done before in [14], there is a gap of knowledge on how these indices relate to the adverse effects of SH [15]. Immunity levels for tests with SH are available in IEC 61000-4-19; they are not directly linked to adverse effects. Researchers that performed immunity tests on electrical appliances have reported flicker and audible noise caused by SH [16]. 2) The relation between the characteristics of SH and their impact on end-user equipment has not been formulated. Some problems related to SH involve, e.g., heating in [16,17] while others involve communication protocol issues [5]. Different features of the SH distortion are responsible for different types of impact. Further work is necessary to understand the morphology of the interferences due to SH [15]; research is ongoing [17,18]. 3) Recommended practices for assessing levels of SH, in the context of problem diagnosis associated with this type of distortion, have not been developed. The lack of agreement on how to diagnose a problem provoked by SH leaves reported issues to remain anecdotal [19,20].

The need for methods to evaluate SH as a function of the incidents they cause arises. This article pioneers the study of the characteristics of SH associated with their effects and the formulation of recommended practices to quantify and evaluate the severity of those effects. The aim of this work is to contribute to the settlement of generally accepted methods for: 1) quantifying the impact of SH; 2) the diagnosis of problems related to SH.

Four interferences caused by SH are considered in this article: audible noise, light flicker, unwanted tripping of LV residual current devices (RCDs), and MV cable termination failure. Two research questions are addressed: 1) What are the relevant attributes of SH associated with each interference phenomenon? By identifying these relevant attributes, input information for the diagnosis of problems existing in an electrical system is obtained. 2) How can the negative effects of SH be assessed by observing the characteristics of SH that predetermine those effects?

The approach and methods employed to answer the stated questions are explained next. A literature review is performed to build the theoretical background of each interference phenomenon. Modelling and simulation, or experiments are performed to replicate the phenomenon. Simple methods or formulas for the evaluation of SH related to each interference are developed. Examples of the application of these methods are presented.

This article contributes to filling the gaps stated above so: 1) The characteristics of SH relevant for each interference are identified. 2) Formulas or methods to quantify the effects of SH on equipment are introduced. 3) Along with appropriate measurements and analysis of the data, identifying the inherence of the problem, contributes to the diagnostic of an incident as caused by SH. Problems caused by SH can thus be diagnosed as such. This will encourage research work on mitigation strategies to move forward. 4) It constitutes a guide on the analysis of SH and diagnosis of problems by means of the evaluation of their characteristics. 5) It constitutes input for standardization committees in charge of defining emission limits in LV and MV and compatibility levels in MV networks.

In the following sections, the four phenomena associated with SH are studied individually. The paper concludes with a summary of recommendations for the evaluation, and a discussion on the attained results.

2. Audible noise

2.1. Reported cases

Audible noise is one of the "most common effects" of SH [5]. Cases of audible noise due to SH have been reported both in literature and unofficially as customers' experiences. Table 1 summarizes some reports of SH causing audible noise found in the literature.

From Table 1, it is seen that SH voltages as low as 0.6 V (0.3 % where the nominal supply voltage is 230 V) can cause audible noise. Except for the case in [21], the SH voltages presented in Table 1 are below the immunity levels in IEC 61000-4-19. A device's compliance with IEC 61000-4-19 does not guarantee its immunity to audible noise due to SH. The latter has been concluded also by other researchers [16].

It is recognized in IEC 61000-2-2, that audible noise can be caused by voltages of at least 0.5 % of the nominal voltage and with frequencies between 1 and 9 kHz.

2.2. Hearing ranges

Human beings can hear frequencies between 20 Hz and 20 kHz. The

Table 1
Some reported cases of audible noise.

Affected equipment	Emitting device(s)	Characteristics of the SH causing the acoustic noise and/or further details
Induction stove	Converter of a variable speed drive of a CNC mill	Components around 5 and 10 kHz with magnitudes 0.6 and 1.7 V (values for the highest 5 Hz-frequency band around 5 and 10 kHz, respectively). The spectrum of the sound revealed a sound pressure level (SPL) of 54 and 43 dB corresponding with the 5 and 10 kHz components [5].
Household devices, bell transformers	Inverter of a hydropower plant	12 kHz component [5]
Electric vehicle (EV)	EVs	Several electric vehicles of one type were charging simultaneously. The affected devices are the emitting devices themselves. The cause is hypothesized to be beating of frequencies very close to each other: switching frequencies of devices of the same type [5].
Office equipment	Two UPS systems	4 kHz switching frequency varying between 0.2 and 0.8 V. The authors of [5] attain beating to the cause of the noise.
Office equipment	Converter-fed pump	12 kHz recurrent oscillations with a magnitude of 40 % of the fundamental voltage at the LV side. The emitting device was connected to the same MV busbar as the building's installation that supplied the affected devices [21].
Television set	Converter in a CNC mill	Components around 8 kHz with a magnitude of 3 V (value for the highest 200-Hz frequency band around 8 kHz) [49].
Household devices	Electric vehicle	10 kHz switching frequency with a magnitude of 0.6 A at the input current of the EV (value for the highest 200 Hz-frequency band around 10 kHz). The frequency of the acoustic noise matched the onboard charger's switching frequency [44].
Induction stove	Heat pump	Several customers complained about buzzing noise in repeated occasions [44].

human hearing response is not linear with respect to the sound pressure level (SPL), and it is most sensitive between 1 kHz and 7 kHz [22]. Factors such as age, previous exposure to high SPL and ear health affect hearing sensitivity [22]. Children can hear frequencies higher than 16 kHz moderately well. The human hearing response to sound pressure is represented by the equal-loudness-level contours available in ISO 226 [22]. A contour is a curve in the SPL vs. frequency plane connecting points whose coordinates represent pure tones judged to be equally loud for a human [22]. The contour at the threshold of hearing in humans is presented in Fig. 1(a). It represents the "level of a sound at which, under specified conditions, a person gives 50 % of correct detection responses on repeated trials" [22].

2.3. State-of-the-art of the research

The acoustic noise generated by electronic devices exposed to SH is due to electromechanical effects on capacitors and coils, e.g., magnetostriction and inverse piezoelectric effect. They can cause mechanical forces that lead to mechanical oscillations. The properties of the audible noise depend on design parameters, e.g., the size of the oscillating surface and the availability of transmission paths to other parts with the ability of vibration [18].

According to the results of the measurement campaign on 103 mass-market end-user equipment [18], levels of acoustic noise created by devices exposed to SH can be as high as 40 dB(A) (A-weighted SPL). About 16 % of the equipment had sound emission that can be disturbing for humans depending on their surroundings. About 12 % of equipment emitted noise reported to be almost always recognized [18]. About 5 % of the devices emitted sound above 32 dB(A); exposure to these has biological effects on humans during their sleep [23].

The tests in [18] revealed that the frequency of the sound coincides with the applied SH frequency. A linear increase in the amplitude of the applied voltage leads to an approximately linear increase in SPL (in dB (A)) but this relation was not studied in detail. The experiments also show that the relation between the magnitude of SH and the SPL depends on the applied frequency. Applying 2 V at 2 kHz and 10 kHz would lead to different SPL depending on the characteristics of the resonating mechanical system. The operation mode of the device subjected to SH voltages has a significant influence on its sound emission. In this sense, it is not possible to generalize the resonance characteristic for all devices.

In another study [16], 55 household devices were exposed to SH adjusted to the immunity levels. Approximately half of the tested devices produced audible noise. Single-frequency SH resulted in more audible noise cases than a band of SH with equivalent rms value. Inductive devices were not affected. Series resonance at the input impedance of the device was suspected to define the emission of audible noise [16].

2.4. Experimental case study

An experiment is performed to replicate the correlation between the amplitude of SH and SPL that was hinted at in [18]. Abstracting this relation contributes to the formulation of indications for SH relative to audible noise.

The device under test is a computer's power supply unit. The device is exposed to pulses of SH superimposed to the mains' voltage. The characteristics of the injected SH are programmed and generated with a controllable power supply. A pattern of SH pulses and pauses with a duration of 3 s each is used. Every pulse carries SH with a different amplitude. A series of 3 pulses are programmed. This test was performed for SH frequencies: 6, 8, and 12 kHz. The sound emitted by the device is measured with a microphone whose output (in V) is proportional to the sound pressure (in Pa). The results are shown in Fig. 2.

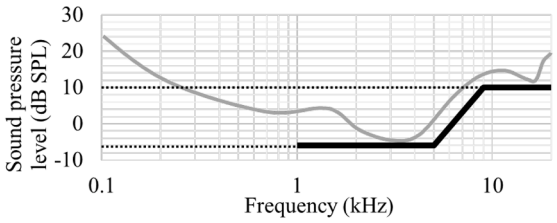
Fig. 2 (a) confirms that higher SH amplitude leads to higher sound pressure. On a shorter scale (100 ms), modulation of the SH component can be observed in Fig. 2(b) for the test with 8 kHz SH frequency. The modulation frequency is twice the mains' nominal frequency; a similar phenomenon is reported in [18]. It is seen in Fig. 2(b), that the sound pressure follows the SH voltage pattern: the highest sound pressure coincides with the highest SH magnitude.

The quantification of the results is made in the time domain. Recorded audio, voltage and current signals are filtered over an 800 Hz-width band centered at the studied frequencies. Only intervals where the amplitude of SH is the highest are quantified; a threshold of amplitude is defined above which the rms values of the signals are calculated. The results are summarized in Fig. 2(c). In Fig. 2(c), the sound pressure increases linearly with the voltage, with a different slope depending on the frequency. The current and the sound slopes coincide for 8 kHz and 12 kHz; an increase in current leads to an equivalent increase in sound pressure for those frequencies. The results at 6 kHz do not support the latter; further research with various devices can contribute to defining a more detailed model. The magnitude of the input impedance of the device is: 9, 6 and 2 Ω, at 6, 8 and 12 kHz, respectively.

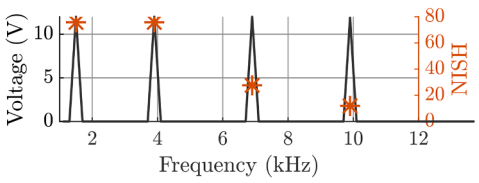
2.5. Recommendations for the evaluation of SH

The authors' recommendation for the evaluation of the severity of SH voltages in the context of audible noise is to weigh SH components depending on their frequency. The frequency-dependency of the SPL of noise caused by SH cannot be generalized for all devices. It is recommended that the weighting be based on the human hearing characteristic (Fig. 1(a)), which is internationally accepted and has been rigorously studied [22].

To develop a weighting formula, the contour at the threshold of hearing is taken as reference (Fig. 1(a)). The range of 1–20 kHz is considered. In Fig. 1(a), a simplification of the contour between 1 kHz and 20 kHz (black solid line) is shown. This curve can be defined mathematically in the plane by a piecewise function in three pieces. The

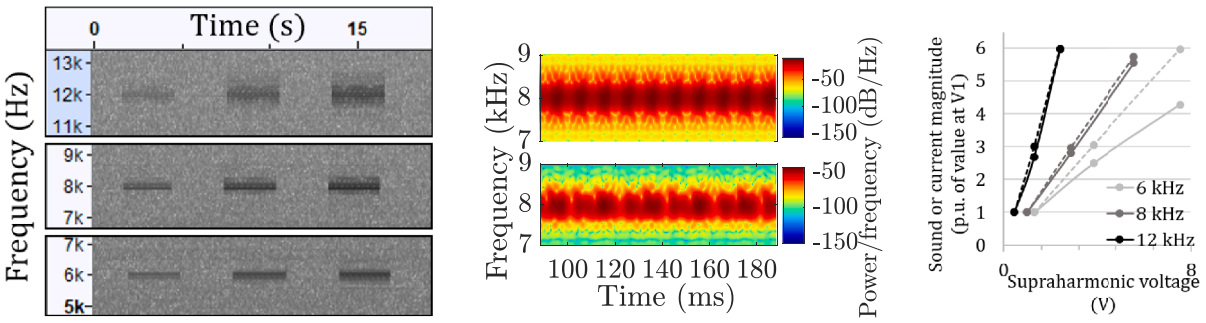


(a) Equal-loudness contour at the threshold of hearing [22], the black solid line shows a simplified curve in the range 1 to 20 kHz



(b) Example of the use of NISH (*). Voltage magnitudes according to IEC 61000-4-19.

Fig. 1. Background and application of the developed formula.



(a) Sound spectrogram. Darker colors mean higher sound pressure. Top, middle and bottom figures correspond to the test with 12, 8 and 6 kHz SH frequency, respectively. SH voltage V1, V2, V3 from left to right. V1<V2<V3.

(b) Comparison of the spectrogram of the voltage (top) and the recorded audio signal (bottom). Results for the test with 8 kHz SH with the highest tested amplitude (5.6 V).

(c) Sound pressure (continuous line) and current (dashed line) vs. injected SH voltage.

Fig. 2. Experiment results: relation between SH voltage and sound pressure.

coordinates defining this piecewise function are (1 kHz, -6 dB), (5 kHz, -6 dB), (9 kHz, 10 dB), and (20 kHz, 10 dB). A weighting coefficient, η, dependent on the frequency is built up based on the simplified contour (Fig. 1(a)) mirrored with respect to the horizontal axis. This is because a weighting formula should give more weight to SH at frequencies where the human hearing is more sensitive. E.g., according to [22], a 20 dB tone created by a SH component at 2 kHz will be perceived louder by a person than a 20 dB tone at 10 kHz. η is defined considering that there is a ratio of 16 dB (6.3 p.u. on a linear scale) between the SPL at 20 kHz and the SPL at 1 kHz for the same perceived loudness (as in the simplified contour in Fig. 1(a)). η is defined in three pieces as in (1):

$$\eta = \begin{cases} 6.3 & \text{if } 1 \le f \le 5 \\ 10^{\left(-\frac{16}{20\log_{10}(9/5)}\log_{10}(f/5)+(16/20)\right)} & \text{if } 5 < f \le 9 \\ 1 & \text{if } 9 < f \le 20, \end{cases} \tag{1}$$

f is the frequency of the SH component in kHz.

In the following, it is assumed that the sound pressure (in Pa) of a tone created by a SH component is proportional to the magnitude of the SH voltage (from Fig. 2(c)). A noise severity indicator for SH (NISH) is formulated in (2) similarly as the telephone influence factor (TIF) in IEEE Std. 519. The spectrum of the voltage signal to evaluate is used. A spectrum with 200 Hz-bands as the result from the method recommended by IEC 61000-4-7 (though extending from 1 kHz to 20 kHz) is suitable since η is defined in bands of minimum 4 kHz width.

$$\text{NISH} = \sqrt{\sum_{B=1}^{95}\left(\eta\frac{V_B}{Z_B}\right)^2}, \tag{2}$$

V_B is the voltage magnitude (rms) corresponding to the band B. Z_B is the magnitude of the input impedance of the affected device; this parameter cannot be defined deterministically at the moment. f_B is the central frequency of the band B and replaces f in (1) for the calculation of η.

NISH serves to compare several SH spectra or components with frequencies between 1 kHz and 20 kHz. NISH benchmarks and gives an indication of the SH distortion's potential to cause audible noise. The four synthetic SH voltage components shown in Fig. 1(b) are taken as examples. The frequencies of the components in Fig. 1(b) are chosen to exemplify the difference between their NISH. For illustrative purposes, Z_B is assumed 1 Ω and constant in frequency.

3. Cable termination failure

3.1. Reported cases

High levels of voltage distortion in the SH range might lead to failure of the insulation in cable terminations in MV installations [24]. High levels of SH are likely to be found in MV networks due to high-frequency system resonances [25].

Failure of cable terminations of the resistive stress-grading type occurred during the commissioning stage of the Eagle Pass installation, which connects the U.S. and the Mexican transmission grids. An investigation revealed voltage distortion with a frequency of 12.4 kHz and amplitude of up to 40 % of the fundamental voltage [24]. This distortion was emitted by the voltage source converter (VSC) and was amplified by resonance. Two additional cases where SH were suspected to be the cause of failure of resistive stress-grading terminations were discussed with different industrial agents; none of the cases have been published. The common solution for all reported cases was replacing the terminations with ones with geometric stress-grading, which signifies an increase in cost and size [24,25].

3.2. Understanding the phenomenon

The function of the stress-grading layer in a cable termination is to control the high electric stress at the edge of the semiconductive screen. The stress-grading layer is designed so that the electric field distributes uniformly over it. As exposed to frequencies higher than the rated power system frequency, the electric field distribution distorts so that high-stress regions are created at the border between the stress-grading layer and the semiconductive screen. High power losses happen in high-stress regions, which in turn create the hotspots on the surface of the cable termination that accelerate the degradation [25].

3.3. State-of-the-art of the research

In [26], six MV cable terminations with non-linear stress grading were subjected to an accelerated aging test; for three of them, SH were superimposed to the power frequency voltage. Two of the latter failed after about five months; the four remaining terminations did not fail during this period. The frequency of the SH distortion was set to 7.2 kHz and its peak-to-peak value, to 15 % of the power frequency voltage. In

[27] a risk indicator for SH and cable termination failure was introduced. The indicator was based on a model of the cable termination [27], and previous theoretical [28] and experimental research [24,29]. The cable termination failure case reported in [24] was used as a reference for the risk indicator.

3.3.1. A simple method for risk evaluation

The attributes of the SH voltage that give information about the power losses on the cable termination are magnitude, V_{SH}, and frequency, f_{SH}. The higher the magnitude and frequency of SH the higher the probability of insulation failure in cable terminations [24]. The model in [27] considers that the heating in the cable termination depends linearly on the frequency of the voltage applied and proportionally with the square of the magnitude of the voltage. A simple evaluation method was introduced in [27] to decide whether given levels and frequencies of SH represent a risk to cable terminations.

The evaluation uses a frequency-domain approach and is made in two steps [27]. The decision is made based on the worst result obtained in each step. This is due to the high variability of characteristics of SH spectra. Some SH have a defined, fixed and recognizable frequency; some extend over a band of frequencies. The difficulty is in finding a reliable representation of the magnitude of the SH (developing such a method is out of the scope of this paper). The two-step evaluation proposed in [27] aimed to cover different scenarios.

Assuming that a frequency spectrum of the voltage is available, with 2 kHz grouping bands from 2 up to 150 kHz, the two-step evaluation is described below. An example of the application of this evaluation was presented in [27].

Compare the spectrum to the risk areas presented in Fig. 3 The SH voltage, V_{SH}, can be expressed as a ratio of the rated voltage, V_{rated}. m gives an indication of the similarity of the evaluated case to the reference case (Eagle Pass). $m = 1$ means that the failure at Eagle Pass occurred under equivalent $\frac{V_{SH}}{V_{rated}}$ and f_{SH} conditions as the evaluated case. The areas originate from a simplification of the Eagle Pass failure condition. The specifications of these areas and further details can be found in [27].

The power loss summation (based on [30]) is considered by the calculation [27]:

$$\mathbf{Q_{SH}} = \sum_{B=1}^{74} \frac{V_B^2 f_B}{V_{rated}^2 f_{rated}}, \tag{3}$$

where V_B and f_B are the voltage magnitude and central frequency of the frequency band, B. In the Eagle Pass case $Q_{SH-EPass} = 20$. The Eagle Pass case serves as a reference and, taking a conservative stance, $m = 0.25$, the risk of failure threshold is defined: $m \cdot Q_{SH-EPass} = 5$.

3.4. Recommendations for the evaluation

The time-wise exposition of the cable termination to SH is relevant. Along with the summary presented in [27], the experiments carried out in [26] give useful information about this matter. In this experiment, the SH were not applied continuously during the whole period of the fundamental voltage but in short pulses every 1.3 ms. Two cable terminations failed under these circumstances and one of them showed a significant increase in the dissipation factor. The latter suggests that, even when the pulses were short and discontinuous, the exposition of the cable terminations to repetitive stress caused by SH during long periods (months) leads to premature aging. Methods for calculating the loss-of-life of cable insulation use Arrhenius equation as well as the current loading pattern over time [31]. This reasoning can be applied to SH in future works. $\mathbf{Q_{SH}}$ can be evaluated statistically by considering the 99% percentile values calculated every 10 minutes.

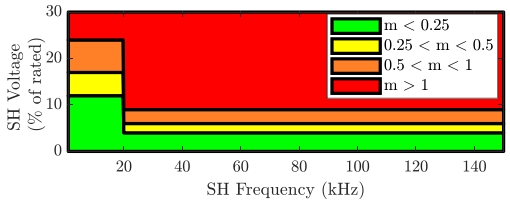


Fig. 3. Risk areas based on m and Eagle Pass case [27].

4. Tripping of LV residual current devices

4.1. Reported cases

Cases of unwanted tripping of LV residual current devices have been reported in several countries [3,19,32–34]. Although SH are the suspected cause, no systematic study has been made to confirm this premise and to understand the mechanism of the incidents regarding the device.

In [3], earth-leakage protective devices' sensitivity provoked the unwanted interruption of electric vehicles (EV) charging. In one case [35], the cause was a ventilation fan that caused high common-mode voltage in the SH range. Ref. [32] reported unwanted tripping of RCDs during EV charging in an EV test laboratory; frequency beating of SH was the suspected cause. Users of EV reported interruptions in the charging process when high levels of SH on the neutral-to-protective earth (N-PE) voltage were present [19]. Communication modems had several communication failures when powered by an uninterruptible power supply (UPS); high levels of SH were found on the phase-neutral and phase-ground voltages [33]. Liew in [34] reported the nuisance tripping of a residual current circuit breaker (RCCB). Measurements revealed 28 kHz components in leakage current with 27 mA peak-to-peak value.

4.2. State-of-the-art of the research

Ref. [36] described the effect of higher-order harmonics on the internal elements of the RCD as "rather complicated and controversial." In some cases, it is possible to speak of the increase of probability of false tripping; in other cases, about the danger of RCD malfunction ("blinding"). Research on the influence of harmonics on RCDs has been performed before [37]. SH have not been studied to that extent. A common conclusion in the literature is that the waveform of the residual current has an impact on the sensitivity of the RCD. A pointed-topped waveform is more prone to cause unwanted tripping than a flat-topped one. It is known that the third harmonic has a strong influence on the waveform shape of the residual current. A phase-angle of the third harmonic in the residual current close to 180° is the worst case [37].

As per manufacturers of RCDs [38,39], one of the most frequent causes of unwanted tripping of RCDs is the leakage of currents to protective earth (PE) through interference filters and stray capacitances in circuits where components up to 20 kHz are present; the earth-leakage current increases with the frequency. Ref. [40], shows the propagation of SH to ground through electromagnetic interference (EMI) filters. Measurements verified that the SH injected by a device propagate to the PE of an adjacent device through its EMI filter, which causes the rms value of the PE current to increase. According to [41], the leakage current, in a household environment, is composed mainly of SH. Nowadays, there are various types of RCDs in the market. Type Bfq and type B+ work with an adapted tripping curve up to 20 kHz. Type A and AC are specified for alternating currents only at 50/60 Hz.

4.3. Understanding the phenomenon

The mechanism in which SH would affect RCDs would be similar to the one for harmonics. To understand the functioning of RCDs under SH, a simple RCD with an electromagnetic relay is considered. This RCD consists of two main parts: the summation current transformer and the

trip relay with a permanent magnet. The summation transformer performs a sum of the instantaneous values of the three phases currents and the neutral current. The residual current (I_Δ) after that summation is considered to be the earth-leakage current. The relay compares that residual current against a given reference and acts once it is surpassed.

An RCD is characterized by its rated residual operating current $I_{\Delta n}$ (typically 30 mA in an LV installation), and its residual non-tripping current, $I_{\Delta n0}$. The former is the effective value of the residual current when the RCD must trip. The latter is the maximum current at which the circuit breaker will not trip and it corresponds to 50% of $I_{\Delta n}$.

The hypothesis of how SH impact the functioning of RCDs is split up into two parts: 1) that EMI filtering increases the probability of RCD tripping (type AC and A); 2) the characteristics of SH have an impact on the sensitivity of the RCD. Notice that statement 1) has to do with the characteristics of the installation, and statement 2) has to do with the characteristics of the RCD. Both are related to the characteristics of the SH distortion.

4.3.1. EMI filtering and SH

The higher the frequency and the amplitude of the SH the higher the probability of unwanted tripping. This is due to the capacitive characteristic of the couplings between phases or neutral and earth. The higher the propagation of SH to earth, the higher the residual current, I_Δ, measured by the current summation transformer in an RCD.

4.3.2. Propagation of SH through the RCD

Consider the simple model of an RCD presented in Fig. 4(a). This case study is merely for exemplification purposes and aims to explain the operation of the RCD under the presence of SH at different frequencies. The circuit represents the summation current transformer in which the primary current corresponds to a given residual current I_Δ. The secondary current flows through the electromagnet of the trip relay (modeled as an inductance). The trip relay is fitted with a permanent magnet. Once the secondary current is conducted into the electromagnet, the attraction force of the permanent magnet is weakened. If the resulting magnetic force is lesser than a given threshold, the relay will trip. This model is a simplified version of that one studied in [37].

In Fig. 4(a), the interest is in the primary and the secondary current, and the resulting magnetic force in the relay. Fig. 4(b) shows the results of the simulation for a synthetic input current with no third harmonic. A threshold is defined for demonstrative purposes. As long as the magnetic force is higher than the threshold, the RCD will not trip. Fig. 4(c) shows the results of the simulation when the third harmonic is present on the residual current. The third harmonic for this signal has a magnitude of 13 mA rms and a 92° phase-angle. The waveform changes and the peak of the current will cause the RCD to trip.

Fig. 4 (c) shows the results of the simulation when SH with 13 mA magnitude and 2 kHz frequency are superimposed to the reference residual current. The magnetic force crosses the threshold so the RCD will trip. A current of the same magnitude but with an increased frequency will result in a lower peak-to-peak value of the magnetic force. The same 13 mA at 2 kHz and 20 kHz at the primary side will end up with different amplitudes at the secondary side due to the frequency-dependent propagation through the summation transformer.

The higher the frequency of the SH is, the lower their propagation to the electromagnet of the relay. The magnitude of the SH that will make the RCD trip is approximately the same as for the magnitude of the third harmonic with 92° phase-angle.

4.4. Evaluation method and discussion

The relative crest factor [42] of the residual current, $CF_{r\Delta}$, as defined in (4) characterizes the shape of the signal and its potential to cause RCD unwanted tripping. $I_{\Delta\max}$ is the maximum value of the residual current and $I_{\Delta\mathrm{rms}}$, its rms value. $CF_{r\Delta} = 1$ for a sine wave and $CF_{r\Delta} > 1$ for a pointed-topped wave. Residual current (primary current) in Fig. 4(b) to 4(c) give $CF_{r\Delta}$ of 1.19, 1.47 and 1.44, respectively.

$$CF_{r\Delta} = \frac{I_{\Delta\max}}{\sqrt{2} I_{\Delta\mathrm{rms}}} \tag{4}$$

The relevant variable for this analysis is the earth-leakage current, which might not be accessible in all cases. An estimation of the earth-leakage current can be done if the phase currents and the neutral currents are available.

The analysis in Section 4.3 leads to conflicting conclusions about how the frequency of SH would impact the unwanted tripping of RCDs. For the system, the higher the frequency of SH the higher the earth-leakage current and the higher the probability of unwanted tripping. On the other hand, the higher the frequency of the SH present in the residual current the higher the energy needed for the RCD to trip. The behavior of the RCD in this regard is highly dependent on its design [37] and at this point, no general conclusions can be drawn on SH impact. Extensive experimental tests are necessary to reach any generalization.

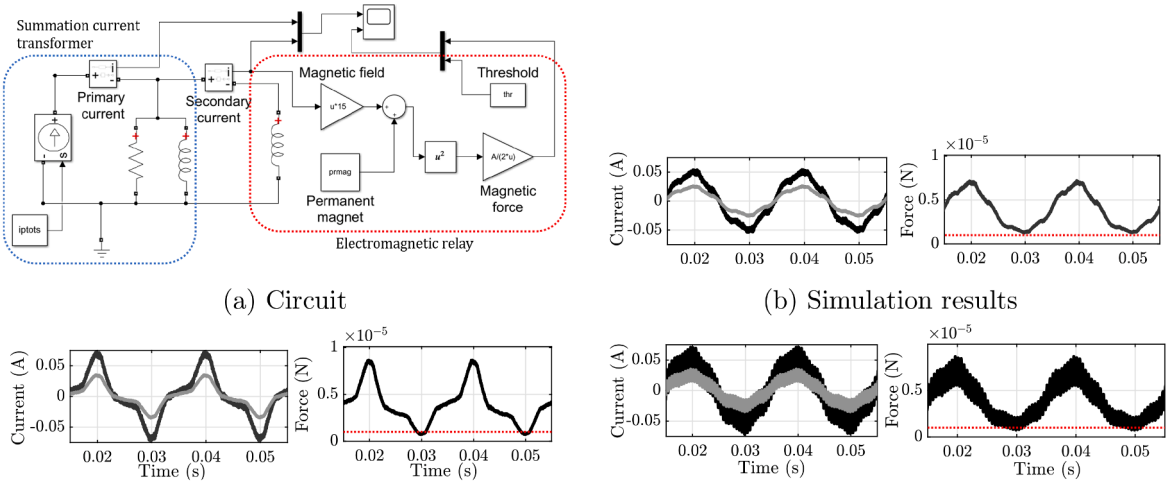


Fig. 4. RCD simulation. Waveform plots – Left: primary current (black), secondary current (gray). Right: magnetic force (solid), threshold (dotted).

5. Light output variations or flicker

5.1. Reported cases

The term ”flicker”, in this section, refers to photometric flicker and describes ”light output variations”. Flicker of LED lamps was observed by a commercial customer in the USA [43]. Investigation of voltage at the location revealed the presence of high-frequency distortion and notches. The distortion showed frequencies between 5 and 10 kHz and amplitudes up to 30 V peak. The distortion was not synchronized with the fundamental voltage; the point-on-wave of the distortion changed with a period of 5 s. Further cases of flicker have been reported in Norway [44], Sweden [19] and USA [20] during the charging of EVs. SH are suspected to be the cause.

5.2. State-of-the-art of the research

It is recognized that LED lamps behave differently from incandescent lamps and that efforts should be made to re-define flicker indicators [45]. The standardized flickermeter defined in IEC 61000-4-15 considers voltage fluctuations with frequencies up to 40 Hz and is based on the response of an incandescent light bulb. SH superimposed on the fundamental voltage can not be perceived by the human eye. A different phenomenon (explained later in this section) is responsible for flicker on LED lamps due to SH and it concerns the functioning of the electronic driver [46].

In [43], five LED lamps were tested under grid voltage superimposed with time- and frequency-varying SH. The point-on-wave of the SH distortion was also time-varying. Two lamps were immune to this SH distortion, one lamp showed a constant decrease in its light output, and two, variations in their light output with a period of approximately 10 s.

In [16], a group of LED and compact fluorescent (CF) lamps were tested under SH with magnitudes adjusted to the immunity levels in IEC 61000-4-19. The flicker assessment was made by visual inspection. Lamps without power factor correction (PFC) stage were not affected by the distortion. Lamps with active PFC flickered when exposed to SH in the range 2 to 20 kHz. Lamps with a capacitor divider topology flickered when exposed to frequencies from 2 up to 95 kHz.

In [46], an LED lamp that consists of a full bridge rectifier with a smoothing capacitor was exposed to a supply voltage superimposed with SH with amplitude 7 V rms at 12.5 kHz. The current at the input of the rectifier and the light output were measured. The interest was in the transition between the conduction and the blocking state of the diodes of the rectifier, which can be seen in the current. It was seen that the SH component forced the diode into blocking/conduction intermittently. The longer this intermittent conduction period was, the stronger the impact of intermittent conduction on the modulation depth of the light intensity output of the lamp. The length of the intermittent conduction period depends on the amplitude and frequency of the voltage SH superimposed to the fundamental voltage. Only SH at the zero-crossing of the current influenced the light intensity-modulation depth. See further details in [46].

Ref. [16,43,46] showed that flicker due to SH is highly dependent on the topology of the lamp. Some lamps are more sensitive than others; some lamps are insensitive to SH.

5.3. Understanding the phenomenon: hypothesis and experimental investigation

The first condition for flicker is intermittent conduction. SH at the zero-crossing of the input current of the LED lamp (causing intermittent conduction) modify the modulation depth of the light output but they do not necessarily cause flicker. The flicker condition meets when SH are not synchronized with the fundamental voltage, i.e., the characteristics of the SH at each current’s zero-crossing are not constant. The latter causes the modulation depth to vary over time which might be sensed as flicker by the human eye. This hypothesis is based on the research presented in [46]. Evidence that supports this hypothesis was found in [43].

One EV user complained about light flicker at home during the charging of the EV. The EV is transported to the laboratory for further investigation. The frequency spectrum and spectrogram of the current of the EV while charging are shown in Fig. 5(a) and 5(b), respectively. In Fig. 5(b), the time-frequency behavior of the SH emission of the EV is represented by the red color. The continuous black line in Fig. 5(b) represents the time domain current waveform which is superimposed on the figure for reference.

The EV emits SH that vary both in frequency and time as seen in Fig. 5(b). As shown in Fig. 5(b), the variation of the frequency of the SH component is not synchronized with the fundamental component’s alternating behavior. It is important to mention that the fundamental current is in phase with the fundamental voltage. The SH variation in Fig. 5(b) is cyclic with 8 cycles per 200 ms, i.e., one cycle lasts 25 ms and the frequency of this variation is 40 Hz.

5.4. Evaluation of the characteristics of SH

In line with the hypothesis in Section 5.3, there are two conditions for flicker caused by SH in sensitive lamps: 1) intermittent conduction, which depends on the amplitude of SH and their point-on-wave in the voltage signal, and the input impedance of the lamp; 2) time-variations of the SH distortion not synchronized with the supply voltage, i.e., time-varying point-on-wave of the SH distortion.

To evaluate whether SH might cause LED lamps to flicker, it is recommended to use time-frequency domain plots and the strategy shown in Fig. 5(b) to look for non-synchronized SH. Namely, inspecting the SH at the zero-crossing of the supply voltage: if they have different characteristics (frequency, magnitude) each half-cycle, then this is a red flag for flicker. The input current of a suspected interfering device can also be inspected, e.g., the current of an EV. The assessment of light output is not covered in this paper. For this matter, refer to IEC 61547-1.

6. Summary of investigation and recommendations

This section summarizes the recommendations and the proposed methods for the evaluation of SH relative to the four studied interferences. The methods are presented in the form of flowcharts that should be read from top to bottom and from left to right. The recommended actions to take on each case are explained afterward.

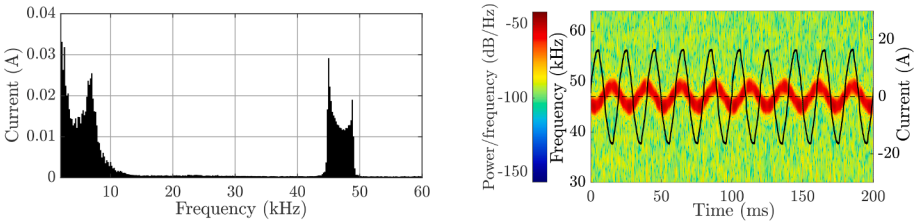
6.1. Processing of data

Proper treatment of the quantification noise is important before the calculation of the indicators. Including quantification noise can affect the results given by the indicators. This process is out of the scope of this paper. A perspective of gap-less measurements is considered here.

6.2. Audible noise

- The existing immunity levels do not guarantee the absence of audible noise due to SH.
- Frequency of SH defines the frequency of audible noise. Switching frequencies and those whose multiples are between 1 and 20 kHz are susceptible to cause audible noise. A single-frequency component is more susceptible to cause audible noise than a band of SH with equivalent rms value.
- The higher the voltage, the higher the sound pressure of the noise produced. This result is device- and frequency-dependent. The input impedance of the device seemingly defines this dependency. Mostly capacitive devices are affected.

Fig. 6 describes the method for the evaluation of SH to identify red



(a) Spectrum, 200 Hz resolution. (b) Non-synchronized SH.

Fig. 5. Input current of an EV that causes light flicker.

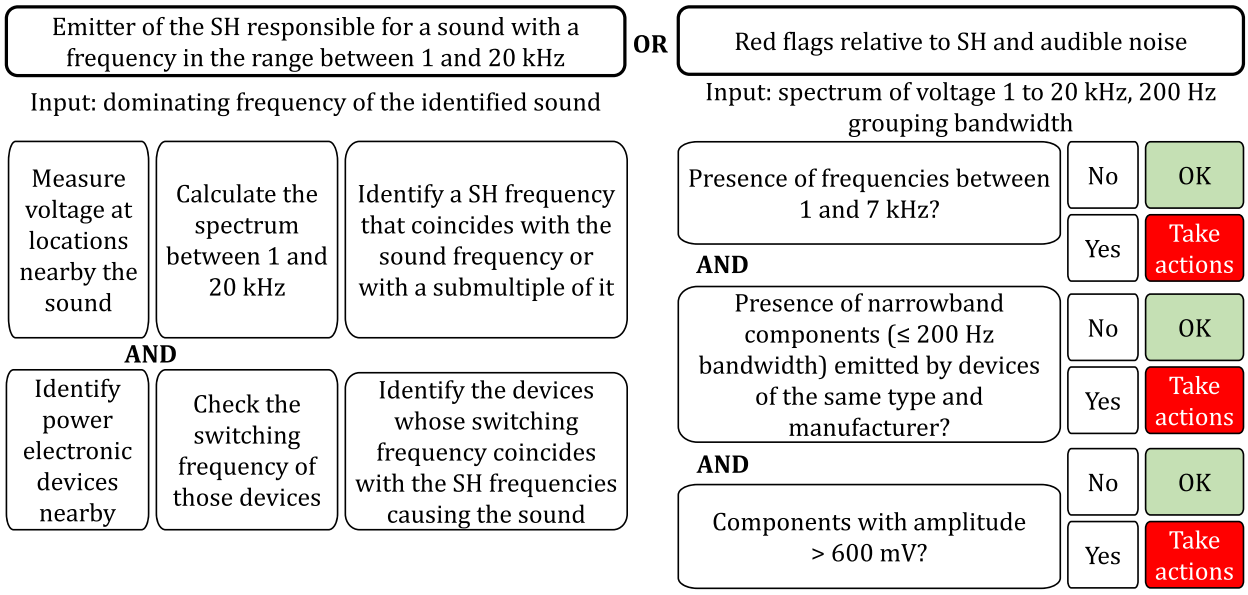


Fig. 6. Flowchart for evaluation of SH related to audible noise.

flags related to audible noise and for finding the source of SH responsible for an identified sound. In case of audible noise caused by SH, EMI filters are a solution. Increasing the electrical distance between the source of the SH and the affected devices is an option that requires further study. The severity of SH voltages related to the risk of them causing audible noise can be quantified using (1) and (2). A reference of SH impedance to model low-voltage devices is needed.

6.3. *Cable terminations*

In cable terminations, resistive stress-grading is a risk factor. There are no records that cable terminations of the geometric type have problems operating under the influence of SH. Higher frequencies and voltage magnitudes represent higher risk. Fig. 7 describes the method for the evaluation of SH to identify red flags related to insulation problems in cable terminations. Eagle Pass case is used as a reference but a more conservative approach is also given. Once a red flag has been

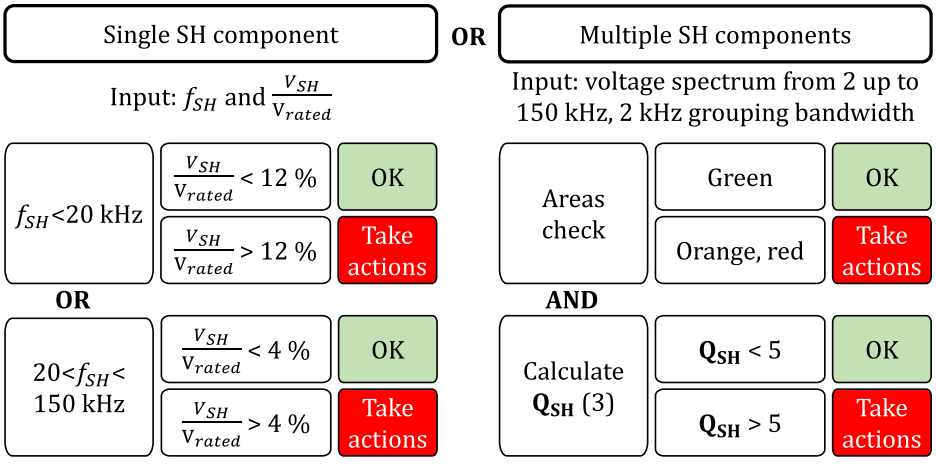


Fig. 7. Flowchart for evaluation of SH related to cable termination failure. Areas and colors refer to Fig. 3.

identified, partial discharge measurements should be paired with dissipation factor measurement for further diagnosis [26].

6.4. Unwanted tripping of RCDs

- The relevant quantity to evaluate is the residual current. It is recommended to keep the residual current under 30% of the nominal tripping current as per IEC 62350.
- Topology-dependent. Types A and AC are more vulnerable.

Fig. 8 describes the method for the characterization of the residual current and the evaluation to identify red flags related to RCDs' unwanted tripping. In case of recurrent unwanted tripping, it is recommended to upgrade the type of RCD to one that is designed to work under high levels of SH, e.g., type B.

6.5. Light flicker

The frequency of SH does not define the frequency of flicker. The amplitude is an influencing factor but the impedance and the topology of the device dominates the condition whether this phenomenon is present. Fig. 9 describes the method for the evaluation of SH to identify red flags related to light flicker on LED lamps. As the phenomenon is dependent on the topology of the LED lamp, this problem can be counteracted by upgrading the lighting equipment to lamps with a different topology.

7. Discussion

The list of negative impacts associated with SH is even longer. Other phenomena such as aging of capacitors, electronic clocks running too fast, interference with PLC have been recognized and are not included in this paper. For some phenomena, an indicator would be rather straightforward, i.e., a curve for acceptable levels of PLC on LV grids already exists in EN 50160 up to 95 kHz. Others require in-depth study, i.e., lifetime degradation of electrolytic capacitors in end-user equipment [17]

7.1. On the prevention and countermeasures to SH-related problems

Two alternatives on how to approach SH-related issues can be considered: either to prevent them or to counteract them once they are reported. Prevention of SH-related issues implies an optimized decision in the context of emission vs. immunity. This is a decision that must be taken by standardization committees in agreement with manufacturers of equipment (both emitters and non-emitters of SH). In its current state, limits for non-intentional SH emission exist for some equipment (LED lamps and induction stoves from 9 up to 150 kHz in EN 55011 and EN 55015, respectively) and for intentional SH emission in EN 50160 (up to 95 kHz). Immunity levels are given in IEC 61000-4-19 but, as explained before, they do not guarantee the absence of SH-related problems. The research conducted here shows that it is crucial to consider the nature of the phenomenon in the development of the following updates of compatibility standards. Once new regulations have been published, the solution can only be applied to equipment in the design stage, e.g., implementing EMI filters and power electronics schemes that modify the spectral characteristics of the emission [47,48]. The second alternative is to focus on counteracting specific problems under the assumption that they will seldom appear. Some options to apply in this case are: installation of filters [24], replacing the affected device [34], modifying the configuration of the local network [11].

8. Conclusion

8.1. Summary of contributions

The methods and formulas proposed to establish a benchmark for measurement, evaluation of SH related to four interferences, and the introduction of a framework for reporting those interferences constitute the most important contributions of this article. These contributions are summarized by Fig. 6 to 9. The innovation of this work lies in the interference-dependent approach to the development of methods for quantifying power quality in the context of SH. Further, this article contributes to the state-of-the-art in research on consequences of SH with the following material:

- An identification of the relevant characteristics of SH related to four interference phenomena: audible noise, cable termination failure, unwanted tripping of RCDs, and light flicker.
- Research shows that the relevant characteristics differ for different phenomena. Performance indicators and limits have been developed where applicable.
- A review of the literature and a summary of reported cases of the four interferences.

8.2. Recommendations for future research

The indicators introduced in this paper are a starting point to quantify the impact of SH. Recommended directions for future work are:

- Immunity tests with a significant sample of devices, registry, and quantification of the effects of SH using the indicators proposed in this article.
- Need the indicators to be more complex to better describe the effects of SH, further research is needed. Increasing the complexity of these methods has drawbacks in terms of implementation since higher computational resources would be required.
- Measurement campaigns on the SH impedance of low-voltage devices.
- Relating the indicators with the propagation of SH, e.g., how far in a distribution feeder can SH cause interferences.

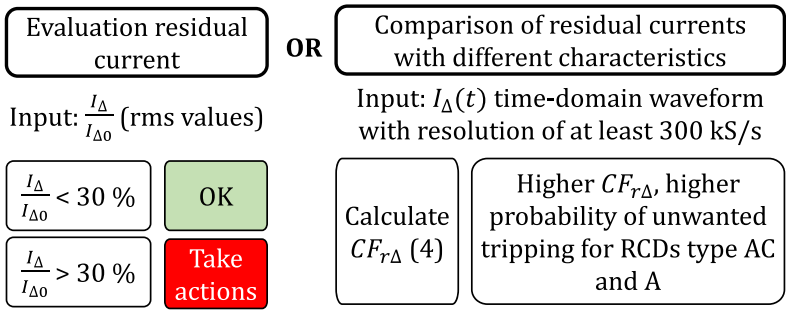


Fig. 8. Flowchart for evaluation of SH related to unwanted tripping of RCDs.

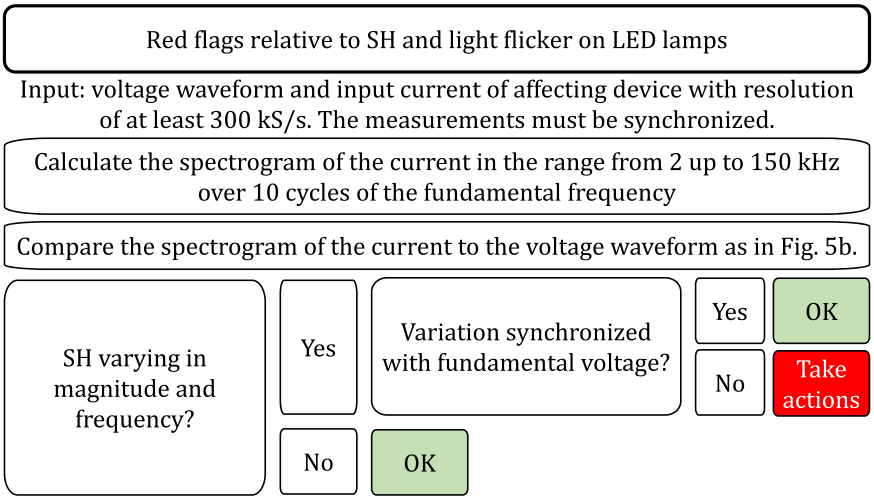


Fig. 9. Flowchart for evaluation of SH related to light flicker.

CRediT authorship contribution statement

Ángela Espín-Delgado: Conceptualization, Formal analysis, Investigation, Methodology, Writing - original draft, Visualization, Validation, Software. **Sarah Rönnberg:** Conceptualization, Methodology, Funding acquisition, Project administration, Resources, Writing - review & editing, Supervision. **Shimi Sudha Letha:** Conceptualization, Methodology, Writing - review & editing. **Math Bollen:** Conceptualization, Methodology, Resources, Funding acquisition, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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